

Oligo Space | Memo

Oligo Space is building the manufacturing infrastructure that will allow enterprises to own application-specific orbital infrastructure without becoming space companies themselves. The company combines AI-native spacecraft design, vertically integrated manufacturing, testing and operations to make customized satellites faster and cheaper than traditional aerospace manufacturers.

The core thesis is simple: launch has become cheap and abundant; satellite manufacturing and mission design are now the bottleneck. Oligo is building the production system for the next phase of the space economy.

Key Investment Highlights

- Space is moving from standardized satellites to application-specific orbital infrastructure.
- Existing manufacturers optimize for either customization or scale, but struggle to deliver both simultaneously.
- Oligo is attacking the manufacturing process itself, using AI to reduce the engineering burden associated with customization.
- Vertical integration gives Oligo control over the design-to-manufacturing loop.
- Every spacecraft creates proprietary design, manufacturing, testing and flight data, creating a compounding closed-loop advantage.
- **The founder is an MIT dropout and a Thiel Fellow and the team combines hands-on manufacturing, aerospace engineering, NASA/space-industry experience and AI/software expertise.**
- **Oligo Space has already raised \$7M to date from investors including Lux Capital, Soma Capital, Alumni Ventures and Kakao Ventures.**

The Opportunity

Space is undergoing a structural transition from government-led, bespoke aerospace programs to commercial, application-specific orbital infrastructure.

Launch costs have fallen dramatically. Falcon 9 reduced the cost of reaching Low Earth Orbit (LEO) by >95% versus the Space Shuttle, while Starship targets sub-\$100/kg economics. Global orbital launches increased from 87 in 2015 to 259 in 2024 and >320 in 2025. Active satellites increased from ~1,000-1,400 a decade ago to >13,000 by late 2025.

The industry is moving from “Can we get something into space?” to “What should we put into space for a specific business?”

The launch and manufacturing represent only 7% of the \$613B global space economy, while 93% flows through applications, services and ground infrastructure. The application layer determines whether infrastructure gets built. Enterprises can either rent data/capacity or own the means of production. A 1997 GAO analysis found that the DoD could meet needs 14x more cheaply by owning capacity rather than leasing it. As costs fall, ownership becomes accessible to enterprises with persistent or proprietary demand.

Oligo addresses the gap between buying generic satellite data/services and building an expensive, slow internal aerospace program.

The Oligo System: Product, Advantage, and Moat

Oligo is effectively building the infrastructure layer that lets an enterprise define a mission and own the spacecraft infrastructure required to execute it. This vertical integration spans the entire lifecycle: business requirement → spacecraft architecture → AI-assisted design → manufacturing → testing → launch → operations → data.

Oligo's AI-driven processes are designed to compress spacecraft development timelines from years to months, while reducing costs from traditional tens of millions of dollars to a fraction of that. Oligo is not an aerospace software company with a manufacturing component; it is building an AI-native physical manufacturing system where software directly improves the economics of producing spacecraft.

Oligo wins by offering a fundamentally better manufacturing system than existing project-based models. Most of the ~120 satellite manufacturers remain engineering-heavy; every new customer requires substantial human involvement in architecture, simulation, component selection, integration, and testing. This creates a linear relationship where more customers require more engineering headcount.

In contrast, Oligo is building Zenith, an AI-native system that turns spacecraft engineering into a repeatable production process. Zenith assists with mission requirements, architectures, design evaluation, and optimization. The real moat is the closed-loop manufacturing data flywheel. Every spacecraft Oligo produces generates data across Design → Simulation → Manufacturing → Testing → Flight → Performance. This feedback loop ensures that each spacecraft makes the next one faster to design, cheaper to manufacture, easier to test, and more reliable.

Over time, Oligo builds a proprietary dataset connecting physical design and manufacturing outcomes to in-orbit performance. The company does not need to be just another satellite manufacturer; it aims to be the production platform that makes customized spacecraft manufacturing economically scalable.

Why Now

Launch is no longer the primary bottleneck. Reusable launch has fundamentally changed space economics and increased orbital access.

Satellite hardware is becoming standardized. The Space Development Agency's order for 126 identical buses is a strong signal that mass production is becoming the expected model.

There is already demand building up visible in the following sectors:

- Financial: Hedge funds spent \$2.8B on alternative data in 2025, including satellite imagery for market pricing.
- Manufacturing: Varda is using microgravity for pharmaceutical manufacturing and has raised \$329M.
- RF: Deloitte is flying nine satellites for RF monitoring.
- Mining / Energy: Satellite data is used for deposit location and methane leak detection.
- Automotive: Geely is investing \$637M in a vehicle-connectivity constellation.
- Agriculture / Consumer: Nestlé, Cargill and Unilever use space for crop-yield forecasting and deforestation verification.
- Insurance: AXA and Munich Re use satellite data to trigger insurance policies.
- Connectivity: Apple has committed \$1.95B to Globalstar for direct satellite connectivity.

Accenture reports that 80% of executives view space technology as vital for growth, but only 18% have integrated it. This suggests significant adoption headroom.

Global telecom operators alone spent \$299.4B in capex in 2024, roughly 38x the total venture funding deployed into space that year. Companies such as Amazon and AT&T operate at a scale where annual capex plans dwarf the space startup ecosystem of the past decade.

Much of the corporate capital is currently flowing into AI infrastructure. As AI capex eventually normalizes, enterprises may seek another strategic frontier for large-scale investment. Space could become a major next corporate spending cycle because orbital data, connectivity and infrastructure provide capabilities that cannot simply be replicated on Earth.

Competitive Landscape

Category	Customization	Repeatability	Enterprise fit
Legacy defense primes	High	Low	Low
Commercial bus builders	Low	High	Medium
Data / service providers	None	High	High
Space infrastructure giants	Medium	High	Medium
Oligo Space	High	High	High

Legacy defense primes: Lockheed Martin and Northrop Grumman offer high customization but low repeatability, with cost-plus structures and multi-year timelines.

Commercial platform builders: Apex, Rocket Lab and Terran Orbital offer repeatability but require missions to fit pre-cataloged hardware.

Data and service operators: Planet and Spire sell generic data feeds, leaving customers exposed to perpetual margins and data-rights limitations.

Infrastructure and utility giants: SpaceX's Starshield adapts standardized buses for government payloads, while Blue Origin's Blue Ring focuses on orbital utility services.

Oligo's target quadrant is customization + repeatability.

Team

Founder(CEO), Jacob Rodriguez brings hands-on manufacturing and aerospace engineering experience. He worked as an aerospace machine-shop technician producing hardware for Virgin Galactic, Rocket Lab, and SpaceX across 3- and 5-axis CNCs. He later studied AeroAstro, physics, and AI at MIT while working six internships at NASA JPL and Goddard. He is also a **Thiel Fellow** and **776 Fellow**.

Assembled a team of veteran engineers that have worked on the world's most advanced flagship missions

70+ YRS OF FLIGHT EXPERIENCE

25+ FLAGSHIP MISSIONS WORKED ON

Andrew Shapiro, PhD
CHIEF TECHNOLOGIST

20+ yrs previous NASA JPL Manager of Technology Formulation and Division Chief Technologist where he spearheaded early-stage innovation and funded seeded over 12 advanced missions across NASA. Lectured at Caltech and UC Irvine. Former Proteus Space Co-Founder/CTO.

Carlos Deccia, PhD
CHIEF MISSION ARCHITECT

Astrodynamics PhD and ML lead engineer. Won a \$8.4M NASA research grant for work in constellation design and remote sensing automation and led AI at Canisius.

Carlyn Lee
SOFTWARE SYSTEMS LEAD

Co-authored the autonomy that won DARPA's SubT Challenge. Led 1000x simulation speedups across JPL's supercomputer cluster, and automated operations for 3 flagship missions.

Samuel Villarreal
HEAD OF FLIGHT SOFTWARE

Owned flight simulations across Mars 2020, Curiosity, and Psyche leading payload definition and flight interface design and ground support equipment testbeds.

Warren Kaye
SENIOR FLIGHT SOFTWARE

10 years of JPL flight software leadership, from leading Lunar Flashlight to spearheading Cassini's Grand Finale maneuver and Psyche. Previous game developer for Call of Duty.

Liliana Reyes
SENIOR AVIONICS

Led the assembly, testing, integration, and delivery of 4 full ESPA spacecraft at Momentous and Proteus Space, achieving the record for fastest delivery of a custom satellite.

Richard Guerrero
MANUFACTURING LEAD

25 years leading the build, test, and integration of flight-qualified hardware and avionics systems at NASA JPL. Air Force veteran led the vertical integration of harnessing at Proteus Space.

Corey Mack
HEAD OF OPERATIONS

20+ yrs JPL operations. Ran ops for a 190-engineer robotics section, managing section logistics, multiple global field tests across flagship missions, lunar rover deployments.

Khari Lobin
MANUFACTURING TECHNICIAN

Automotive manufacturing technician at BMW. Welding, assembly/integration, and precision fabrication experience across automotive, HVAC systems, PC assembly, and IT infrastructure.

Progress So Far

- Established a 14,000 sq. ft. spacecraft production facility in Hawthorne, next to SpaceX.
- Built in-house cleanroom, TVAC, vibration testing, ADCS testing, machining and waterjet capabilities.
- **Built up the capability to produce over 10 custom spacecraft ranging between 100-400kg at their facility in Hawthorne every year.**
- Booked and paid for the launch of its first 100kg satellite, CHIMERA1, in April 2027.
- Acquired customers for CHIMERA1 and CHIMERA2 both scheduled to launch in 2027 with a possible revenue on each CHIMERA ranging from \$15M-\$37M.
- Continued development of Zenith, its AI-enabled spacecraft design automation platform.
- Built automated design, testing and manufacturing workflows that improve with every customer and spacecraft build.

Investment Status

- Oligo Space has raised \$7M to date from investors including **Lux Capital, Soma Capital, Alumni Ventures and Kakao Ventures.**
- The last strategic round was closed at a \$50M valuation.
- The company is currently raising \$10M at a \$100M valuation.

Conclusion

Oligo Space is a bet on the next manufacturing layer of the space economy. Declining launch costs and standardized satellite platforms have created a new bottleneck: how enterprises efficiently design and manufacture spacecraft optimized for their specific applications.

Oligo can win not by competing head-on with every satellite manufacturer, but by changing the economics of customization itself. If it converts spacecraft engineering from a labor-intensive project business into an AI-driven, repeatable manufacturing process, its advantage should compound with every spacecraft it builds.

Bottom line: Oligo offers a differentiated way to invest in the transition from “space as a service” to “space as enterprise-owned infrastructure.”